

Introduction to Modeling for Formal Verification

Today's goals

Module 1

Introduction to Model Checking

Module 2

Modeling

Background

Propositional Logic

Video • 13 min

Predicate Logic

Video • 9 min

Set Theory

Video • 11 min

Foundations of Logic and Set Theory - Key Concepts and Applications

Reading • 1h

Background Quiz

Graded Assignment • 30 min

Transition Systems

Transition Systems: Definitions

Video • 13 min

Deterministic vs Non-Deterministic Systems

Video • 8 min

Executions

Video • 5 min

Modeling Sequential Circuits

Video • 11 min

Principles of Model Checking - Section 2.1, Page 19-29

Reading • 45 min

Transition Systems Quiz

Graded Assignment • 20 min

Program Graphs

Program Graphs: Definitions

Video • 9 min

Program Graphs: Examples

Video • 15 min

From Program Graphs to Transition Systems

Video • 11 min

Principles of Model Checking - Section 2.1, (Page 29-34)

Reading • 30 min

Program Graphs Quiz

Graded Assignment • 15 min

Module 3

Modeling System Composition

Modeling Concurrency Using Transition Systems

Concurrency and Interleaving

Video • 10 min

Concurrency and Interleaving: Examples

Video • 8 min

Principles of Model Checking - Section 2.2.1, (Page 35-39)

Reading • 30 min

Modeling Concurrency Using Transition Systems Quiz

Graded Assignment • 15 min

Modeling Communications Via Shared Variables

Interleaving Program Graphs

Video • 11 min

Critical Actions and Atomicity

Video • 10 min

Peterson's Mutual Exclusion Algorithm

Video • 12 min

Principles of Model Checking - Section 2.2.2, (Page 39-47)

Reading • 30 min

Modeling Communications Via Shared Variables Quiz

Graded Assignment • 20 min

Modeling Communications Via Handshaking

Synchronization and Handshaking

Video • 12 min

Example: Railroad Crossing

Video • 13 min

Principles of Model Checking - Section 2.2.3, (Page 48-52)

Reading • 30 min

Modeling Communications Via Handshaking Quiz

Graded Assignment • 10 min

Synchronous Parallelism

Synchronous Product - Part 1

Video • 10 min

Synchronous Product - Part 2

Video • 10 min

The State Space Explosion Problem

Video • 9 min

Principles of Model Checking - Section 2.2.6, (Page 73-75)

Reading • 20 min

Synchronous Parallelism Quiz

Graded Assignment • 20 min

Foundations of Logic and Set Theory - Key Concepts and Applications



0:09 / 1:33

2x





Textbook - Principles of Model Checking - Appendix A.1, A.3

A.1 introduces notations and symbols used in the book and discussions provided in this course. It also contains some key concepts of set theory. A.3 introduces some key concepts of propositional logic necessary for this course.

First-Order Logic - Wikipedia Page

The Wikipedia page on the first-order logic gives a broader introduction to first-order logic including predicate logic.

Read about first-order logic at the following link: [First-Order Logic - Wikipedia Page](#) (external link; opens in new tab).

Reasoning - Introduction, Predicates and Quantifiers

This online book provides a good introduction to predicates and quantifiers. It also contains exercises for various topics covered in this introduction.

Access section 4.1 of this book at the following link: [Reasoning - Introduction, Predicates and Quantifiers](#) (external link; opens in new tab).

Discrete Mathematics with Applications (5th Edition) - Chapter 2, 3, 6, 8

This textbook by Susanna S. Epp provides comprehensive coverage on various subjects on discrete math. Chapter 2 introduces propositional logic. Chapter 3 introduces predicate logic. Chapter 6 introduces set theory. Chapter 8 introduces relations while Chapter 7 introduces functions, which are a specific form of relations.

The textbook can be found at the following link: [Discrete Mathematics with Applications \(5th Edition\)](#) (external link; opens in new tab).

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